



# GATE ME 90 Days Challenge

Day 2/90 · Practice Set with Detailed Solutions

SUBJECT

**Engineering Mechanics**

TODAY'S TOPIC

**Kinematics & Dynamics**

COUNTDOWN

**109 days to GATE**

**How to use this set:** 15 questions (7 MCQ, 8 NAT) on kinematics and dynamics of particles and rigid bodies in plane motion. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

**Today's free test:** 7:00 PM on our website and app, 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

## Questions

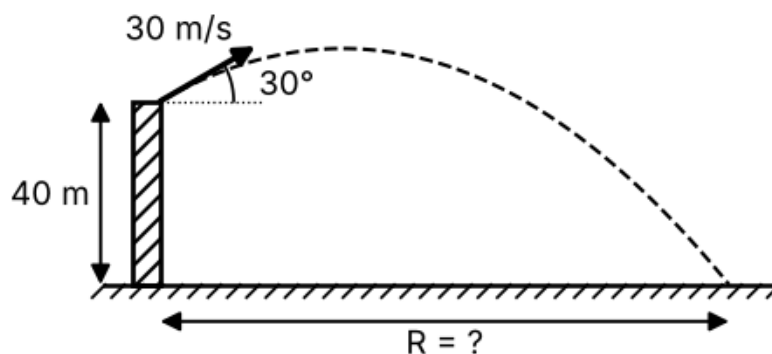
**Q1** (MCQ)

A particle moves along a straight line with  $x = t^3 - 6t^2 + 9t + 2$  (x in m, t in s). The total distance travelled between  $t = 0$  and  $t = 4$  s is

- (A) 4 m (B) 8 m  
(C) 12 m (D) 16 m

**Q2** (NAT)

A ball is projected from the top of a 40 m high tower with a speed of 30 m/s at  $30^\circ$  above the horizontal, as shown. Taking  $g = 9.81 \text{ m/s}^2$ , the horizontal distance from the foot of the tower to the point where the ball hits the ground (in m) is \_\_\_\_\_



**Q3** (MCQ)

A boat moving at 20 m/s switches off its engine. The water resistance gives a retardation  $a = -0.5v$  (a in  $\text{m/s}^2$ , v in  $\text{m/s}$ ). The distance covered while the speed falls to 5 m/s is

- (A) 15 m (B) 27.7 m  
(C) 40 m (D) 30 m

**Q4** (NAT)

A car moves on a circular track of radius 100 m. At an instant its speed is 15 m/s and the speed is increasing at  $2 \text{ m/s}^2$ . The magnitude of the total acceleration of the car at that instant (in  $\text{m/s}^2$ ) is \_\_\_\_\_

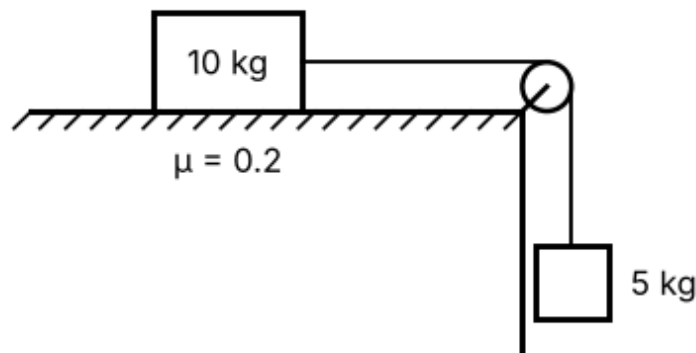
**Q5** (MCQ)

A road curve of radius 120 m is to be banked so that vehicles at 54 km/h need no friction. The required banking angle is

- (A)  $10.8^\circ$  (B)  $5.4^\circ$   
(C)  $24.6^\circ$  (D)  $15.0^\circ$

**Q6** (NAT)

A 10 kg block on a rough horizontal table ( $\mu = 0.2$ ) is connected by a light inextensible string over a smooth light pulley to a hanging 5 kg block, as shown. The tension in the string (in N) is \_\_\_\_\_

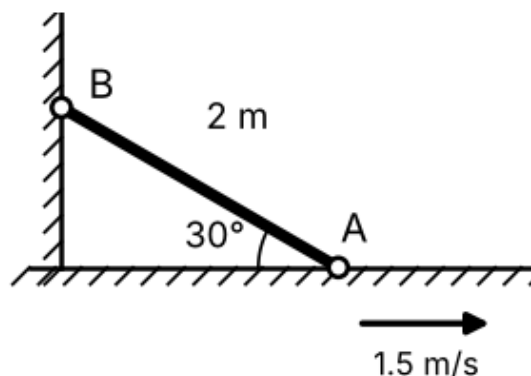
**Q7** (MCQ)

A uniform solid disc rolls without slipping on a horizontal floor. The ratio of its rotational kinetic energy to its total kinetic energy is

- (A)  $1/2$  (B)  $1/3$   
(C)  $2/3$  (D)  $1/4$

**Q8** (NAT)

A rod AB of length 2 m slides with end A on the floor and end B on a vertical wall, as shown. When the rod makes  $30^\circ$  with the floor, A moves away from the wall at 1.5 m/s. The speed of B (in m/s) at this instant is \_\_\_\_\_



**Q9** MCQ

A solid sphere rolls down a  $30^\circ$  incline without slipping, starting from rest. The acceleration of its centre is ( $g = 9.81 \text{ m/s}^2$ )

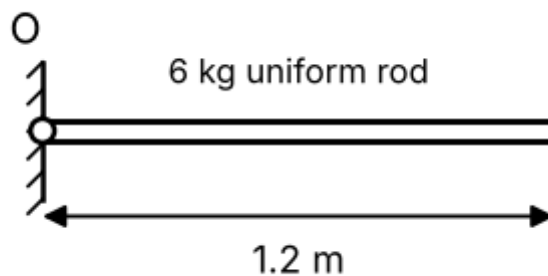
- (A)  $4.905 \text{ m/s}^2$  (B)  $3.270 \text{ m/s}^2$   
(C)  $3.504 \text{ m/s}^2$  (D)  $2.453 \text{ m/s}^2$

**Q10** NAT

The speed of a flywheel falls uniformly from 300 rpm to 120 rpm in 30 s. The number of revolutions it makes in this time is \_\_\_\_\_

**Q11** NAT

A uniform rod of mass 6 kg and length 1.2 m is hinged at end O and held horizontal, as shown. It is released from rest. The vertical reaction at the hinge just after release (in N) is \_\_\_\_\_

**Q12** MCQ

A wheel of radius 0.4 m rolls without slipping; its centre moves at 10 m/s. The speed of a point on the rim at the same height as the centre is

- (A) 10 m/s (B) 20 m/s  
(C) 14.14 m/s (D) 7.07 m/s

**Q13** NAT

A man of mass 70 kg stands on a weighing scale inside a lift. The lift accelerates upward at  $2 \text{ m/s}^2$ . The scale reading (in N) is \_\_\_\_\_

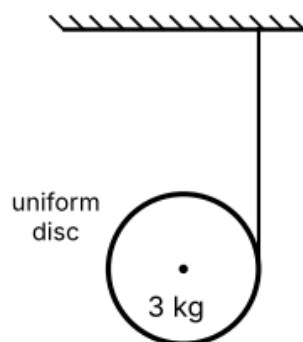
**Q14** MCQ

A small stone tied to a 1 m string is whirled in a vertical circle. The least speed at the lowest point that lets it just complete the circle is

- (A) 3.13 m/s (B) 4.43 m/s  
(C) 6.26 m/s (D) 7.00 m/s

**Q15** (NAT)

A string is wound round a uniform disc of mass 3 kg. The free end of the string is fixed to the ceiling and the disc is released from rest, the string staying vertical, as shown. The acceleration of the centre of the disc (in  $\text{m/s}^2$ ) is \_\_\_\_\_



## Answer Key

| Q   | Type | Answer                                |
|-----|------|---------------------------------------|
| Q1  | MCQ  | (C) 12 m                              |
| Q2  | NAT  | 123.9 (accept 123 to 125)             |
| Q3  | MCQ  | (D) 30 m                              |
| Q4  | NAT  | 3.01 (accept 2.98 to 3.04)            |
| Q5  | MCQ  | (A) $10.8^\circ$                      |
| Q6  | NAT  | 39.24 (accept 39.0 to 39.5)           |
| Q7  | MCQ  | (B) $1/3$                             |
| Q8  | NAT  | 2.598 (accept 2.58 to 2.62), downward |
| Q9  | MCQ  | (C) $3.504 \text{ m/s}^2$             |
| Q10 | NAT  | 105                                   |
| Q11 | NAT  | 14.72 (accept 14.5 to 14.9)           |
| Q12 | MCQ  | (C) $14.14 \text{ m/s}$               |
| Q13 | NAT  | 826.7 (accept 825 to 828)             |
| Q14 | MCQ  | (D) $7.00 \text{ m/s}$                |
| Q15 | NAT  | 6.54 (accept 6.50 to 6.58)            |

## Detailed Solutions

### Q1 Answer: (C) 12 m

**Concept:** Distance is not displacement. Find where the velocity changes sign and add the lengths of each leg.

$v = dx/dt = 3t^2 - 12t + 9 = 3(t - 1)(t - 3) \rightarrow$  the particle turns round at  $t = 1 \text{ s}$  and  $t = 3 \text{ s}$ .

$x(0) = 2 \text{ m}$ ,  $x(1) = 6 \text{ m}$ ,  $x(3) = 2 \text{ m}$ ,  $x(4) = 6 \text{ m}$

Distance =  $|6 - 2| + |2 - 6| + |6 - 2| = 4 + 4 + 4 = \mathbf{12 \text{ m}}$  (displacement is only 4 m).

**Exam tip:** Whenever  $v$  can change sign inside the interval, split the interval at  $v = 0$  before adding.

### Q2 Answer: 123.9 (accept 123 to 125)

**Concept:** Horizontal and vertical motions are independent. Use the vertical motion to find the time of flight, then  $x = u \cos \theta \cdot t$ .

Components:  $u_x = 30 \cos 30^\circ = 25.98 \text{ m/s}$ ,  $u_y = 30 \sin 30^\circ = 15 \text{ m/s}$  (up).

Taking up as positive, the ball lands at  $y = -40 \text{ m}$ :  $-40 = 15t - 4.905t^2 \rightarrow 4.905t^2 - 15t - 40 = 0$   
 $t = [15 + \sqrt{(225 + 784.8)}] / 9.81 = \mathbf{4.768 \text{ s}}$  (the negative root is rejected).

$R = 25.98 \times 4.768 = \mathbf{123.9 \text{ m}}$

**Exam tip:** Do not use  $R = u^2 \sin 2\theta / g$  here. That formula holds only when launch and landing are at the same level.

**Q3 Answer: (D) 30 m**

**Concept:** When  $a$  is given as a function of  $v$  and distance is wanted, write  $a = v dv/dx$ .

$$v dv/dx = -0.5v \rightarrow dv/dx = -0.5$$

$$x = (20 - 5)/0.5 = \mathbf{30 \text{ m}}$$

(The time taken would come from  $dv/dt = -0.5v$ :  $t = 2 \ln 4 = 2.77 \text{ s}$ , a common distractor.)

**Exam tip:**  $a = dv/dt$  gives time;  $a = v dv/dx$  gives distance. Pick the form that matches what is asked.

**Q4 Answer: 3.01 (accept 2.98 to 3.04)**

**Concept:** In curvilinear motion  $a = a_t$  (change of speed) +  $a_n$  (change of direction), and the two are perpendicular.

$$a_n = v^2/r = 15^2/100 = \mathbf{2.25 \text{ m/s}^2} \text{ (towards the centre)}$$

$$a_t = dv/dt = \mathbf{2 \text{ m/s}^2} \text{ (along the path)}$$

$$a = \sqrt{(2.25^2 + 2^2)} = \mathbf{3.01 \text{ m/s}^2}$$

**Exam tip:** Never add  $a_t$  and  $a_n$  as numbers (4.25 is a trap). They are at right angles.

**Q5 Answer: (A) 10.8°**

**Concept:** With no friction, the horizontal component of the normal reaction alone supplies the centripetal force:  $\tan \theta = v^2/(rg)$ .

$$v = 54 \text{ km/h} = 15 \text{ m/s}$$

$$\tan \theta = 15^2 / (120 \times 9.81) = 0.1911 \rightarrow \theta = \mathbf{10.82^\circ} \approx 10.8^\circ$$

**Exam tip:** Always convert km/h to m/s (divide by 3.6) before using  $v^2/r$ .

**Q6 Answer: 39.24 (accept 39.0 to 39.5)**

**Concept:** Write Newton's second law for each block along its own direction of motion; both have the same acceleration  $a$ .

$$\text{Hanging block: } 5g - T = 5a$$

$$\text{Table block: } T - 0.2 \times 10g = 10a$$

$$\text{Adding: } 5g - 2g = 15a \rightarrow a = 3 \times 9.81/15 = \mathbf{1.962 \text{ m/s}^2}$$

$$T = 5(g - a) = 5 \times 7.848 = \mathbf{39.24 \text{ N}}$$

**Exam tip:** First check that motion happens at all: here  $5g = 49 \text{ N}$  is larger than the limiting friction  $2g = 19.6 \text{ N}$ , so it does.

**Q7 Answer: (B) 1/3**

**Concept:** For rolling without slipping  $v = rw$ , so  $KE_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2(1 + k^2/r^2)$ .

$$\text{Disc: } I = \frac{1}{2}mr^2 \rightarrow KE_{\text{rot}} = \frac{1}{2}(\frac{1}{2}mr^2)(v/r)^2 = \frac{1}{4}mv^2$$

$$KE_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{4}mv^2 = \frac{3}{4}mv^2$$

$$\text{Ratio} = (\frac{1}{4})/(\frac{3}{4}) = \mathbf{1/3}$$

**Exam tip:** Ring 1/2, disc 1/3, solid sphere 2/7. The ratio is  $(k^2/r^2)/(1 + k^2/r^2)$ .

**Q8 Answer: 2.598 (accept 2.58 to 2.62), downward**

**Concept:** The instantaneous centre I lies where the perpendiculars to the velocities of A and B meet: vertically above A and horizontally level with B.

IA = height of B =  $2 \sin 30^\circ = 1$  m; IB = horizontal distance of A =  $2 \cos 30^\circ = 1.732$  m

$\omega = v_A/IA = 1.5/1 = 1.5$  rad/s

$v_B = \omega \times IB = 1.5 \times 1.732 = \mathbf{2.598 \text{ m/s}}$  (downward). Equivalently  $v_B = v_A \cot \theta$ .

**Exam tip:** Differentiating  $x^2 + y^2 = L^2$  gives  $x \cdot v_A = y \cdot v_B$ , a quick check on any ladder problem.

**Q9 Answer: (C) 3.504 m/s<sup>2</sup>**

**Concept:** For rolling without slipping on an incline  $a = g \sin \theta / (1 + k^2/r^2)$ .

Solid sphere:  $k^2/r^2 = 2/5 \rightarrow a = (5/7) g \sin \theta$

$a = (5/7) \times 9.81 \times 0.5 = \mathbf{3.504 \text{ m/s}^2}$

The other options are for a smooth slide (4.905), a disc (3.270) and a ring (2.453).

**Exam tip:** The smaller  $k/r$  is, the faster the body rolls down. Solid sphere beats disc beats ring.

**Q10 Answer: 105**

**Concept:** With uniform angular retardation, angle turned = average angular speed  $\times$  time.

Average speed =  $(300 + 120)/2 = 210$  rpm

Time = 30 s = 0.5 min

Revolutions =  $210 \times 0.5 = \mathbf{105}$

**Check:**  $\alpha = (4\pi - 10\pi)/30 = -0.6283 \text{ rad/s}^2$ ;  $\theta = \omega_0 t + \frac{1}{2}\alpha t^2 = 10\pi(30) - \frac{1}{2}(0.6283)(900) = 659.7$  rad = 105 rev ✓

**Exam tip:** Stay in rpm and minutes when you only need revolutions. It avoids  $2\pi$  conversions.

**Q11 Answer: 14.72 (accept 14.5 to 14.9)**

**Concept:** Just after release  $\omega = 0$ , so the centre G has only tangential (vertical) acceleration  $a_G = \alpha \cdot L/2$ .

Moments about O:  $mg(L/2) = (mL^2/3) \alpha \rightarrow \alpha = 3g/(2L) = 3 \times 9.81/2.4 = \mathbf{12.26 \text{ rad/s}^2}$

$a_G = \alpha L/2 = 12.26 \times 0.6 = 7.36 \text{ m/s}^2 = (3/4)g$

Vertical:  $mg - R = m a_G \rightarrow R = mg/4 = 6 \times 9.81/4 = \mathbf{14.72 \text{ N}}$

**Exam tip:**  $R = mg/4$  at the instant of release is a standard result. Memorise it, but know the derivation.

**Q12 Answer: (C) 14.14 m/s**

**Concept:** In rolling, every point moves as if rotating about the contact point (the instantaneous centre) with  $\omega = v/r$ .

$\omega = 10/0.4 = 25 \text{ rad/s}$

Distance of the point from the contact point =  $\sqrt{(0.4^2 + 0.4^2)} = 0.5657 \text{ m}$

$v = 25 \times 0.5657 = \mathbf{14.14 \text{ m/s}} = \sqrt{2} \times v_{\text{centre}}$

**Exam tip:** Top point  $2v$ , centre  $v$ , contact point 0, side points  $\sqrt{2} v$ . The radius of the wheel does not matter.

**Q13 Answer: 826.7 (accept 825 to 828)**

**Concept:** The scale reads the normal reaction  $N$ . Apply Newton's second law to the man.

$$N - mg = ma \rightarrow N = m(g + a)$$

$$N = 70 \times (9.81 + 2) = \mathbf{826.7 \text{ N}}$$

**Exam tip:** Upward acceleration (or downward deceleration) increases the reading; free fall makes it zero.

**Q14 Answer: (D) 7.00 m/s**

**Concept:** At the top the string may just go slack:  $T = 0 \rightarrow mg = mv_{\text{top}}^2/r \rightarrow v_{\text{top}}^2 = gr$ .

Energy between bottom and top (height  $2r$ ):  $v_{\text{bot}}^2 = v_{\text{top}}^2 + 4gr = 5gr$

$$v_{\text{bot}} = \sqrt{(5 \times 9.81 \times 1)} = \mathbf{7.00 \text{ m/s}}$$

**Exam tip:** For a rod (which can push) instead of a string, the top speed can be zero, so  $v_{\text{bot}} = \sqrt{(4gr)}$ .

**Q15 Answer: 6.54 (accept 6.50 to 6.58)**

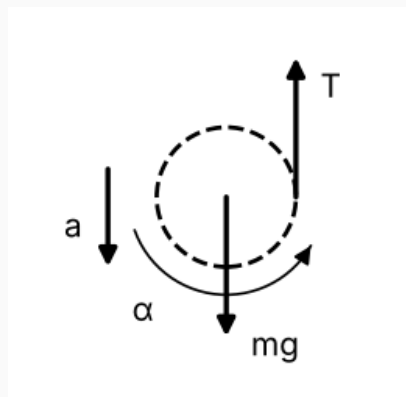
**Concept:** The disc translates and rotates. The string unwinds without slipping, so  $a = r\alpha$ .

Translation:  $mg - T = ma$

Rotation about the centre:  $T r = (\frac{1}{2}mr^2) \alpha = \frac{1}{2}mr(a) \rightarrow T = \frac{1}{2}ma$

$$mg = 1.5 ma \rightarrow a = 2g/3 = \mathbf{6.54 \text{ m/s}^2}, \text{ and } T = mg/3 = \mathbf{9.81 \text{ N}}$$

**Exam tip:** The mass and radius cancel. For a ring the answer would be  $g/2$ , for a solid sphere  $5g/7$ .



## What next?

### Free Day 2 test tonight, 7:00 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with photo, name and marks in the next 8 AM challenge reel.



Website test series



Dig Career Thrust app



Telegram: daily PDFs

**Next, Day 3/90:** Engineering Mathematics, Linear Algebra. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.